Hyperlink Induced Topic Search Algorithm

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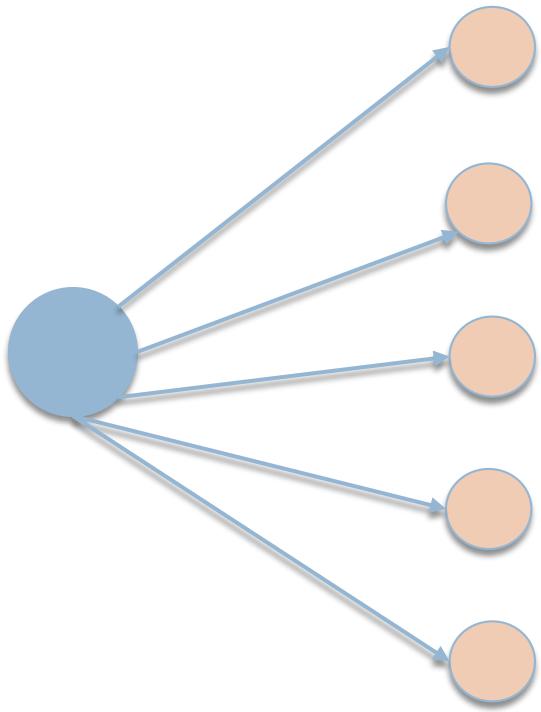
Hyperlink Induced Topic Search Algorithm

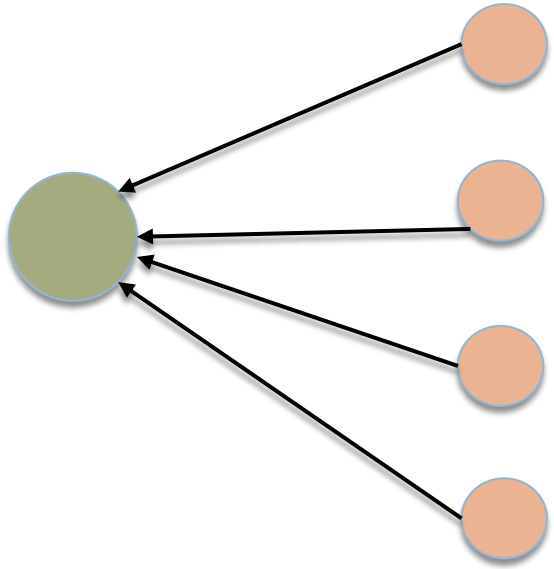
- Hyperlink Induced Topic Search (HITS) Algorithm is based on the concept of Hub nodes and Authority nodes.
- **Hub Nodes:** Pages that act as directories or indices, linking to many authoritative pages. A good hub is a page that points to many good authorities.
- **Authority Nodes:** Pages that contain valuable, high-quality information on a given topic. They are considered important because many other pages (hubs) link to them.
- **Mutual Reinforcement:** The central idea is that a good authority is linked to by many good hubs, and a good hub points to many good authorities. The scores reinforce each other iteratively.

Hyperlink Induced Topic Search Algorithm

- A Hub



- An Authority

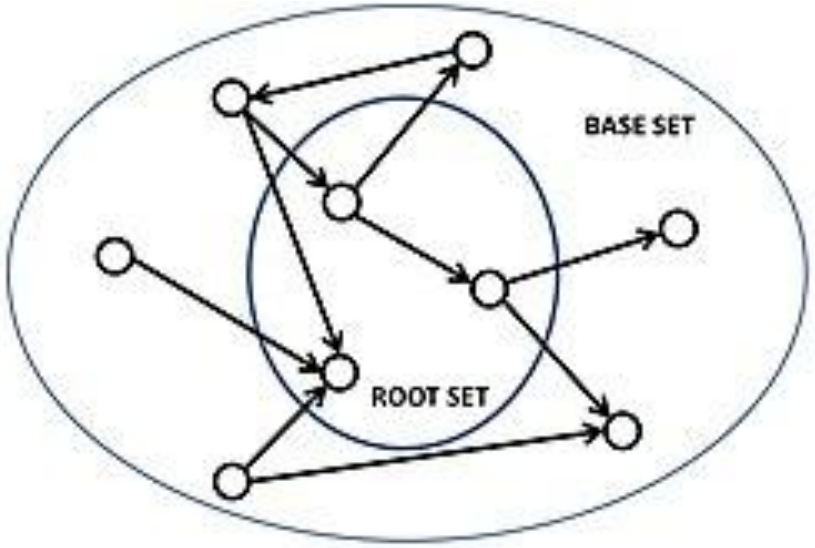


Hyperlink Induced Topic Search Algorithm

The HITS algorithm involves the following steps:

- **Initial Set Construction:** A set of web pages relevant to a specific query is gathered (the "root set"). This set is expanded to include pages linking to or linked from the root set (the "base set").
- **Initialization:** Each page in the base set is assigned an initial authority score and hub score, usually set to 1.
- **Iterative Updates:** The algorithm repeatedly updates the scores in two steps until they converge:
  - **Authority Update:** Each page's new authority score is the sum of the hub scores of all the pages that link to it.
  - **Hub Update:** Each page's new hub score is the sum of the authority scores of all the pages it links to.
- **Normalization:** After each iteration, all hub and authority scores are normalized (e.g., by dividing each score by the square root of the sum of the squares of all scores of the same type (hub or authority)) to prevent the values from growing indefinitely.

Root Set and Base Set



Hyperlink Induced Topic Search Algorithm

Applications:

Besides web search, the HITS algorithm's principles are applied in other network analysis contexts such as:

- Social network analysis to find influential people (hubs) and trusted information sources (authorities).
- Citation networks to identify important academic papers (authorities) and review articles (hubs).
- Recommendation systems.

How to compute hub and authority scores

- Given a broad search query, q, HITS collects a set of pages as follows:
  - It sends the query q to a search engine
  - It then collects t (t = 200 is used in the HITS paper) highest ranked pages. This set is called the root set W.
  - It then grows W by including any page pointed to by a page in W and any page that points to a page in W. This gives a larger set S, base set.
  - HITS works on the pages in S, and assigns every page in S an authority score and a hub score.
  - Let the number of pages in S be n.
  - We use G = (V, E) to denote the hyperlink graph of S.
  - We use L to denote the adjacency matrix of the graph.

How to compute hub and authority scores

- Let the authority score of the page i be a(i), and the hub score of page i be h(i).
- The mutual reinforcing relationship of the two scores is represented as follows:

$$a(i) = \sum_{(j,i)\in E} h(j)$$

$$h(i) = \sum_{(i,j)\in E} a(j)$$

How to compute hub and authority scores

- We use a to denote the column vector with all the authority scores,

$$a = [a(1), a(2), \ldots, a(n)]T, \text{ and}$$

- Use h to denote the column vector with all the hub scores,

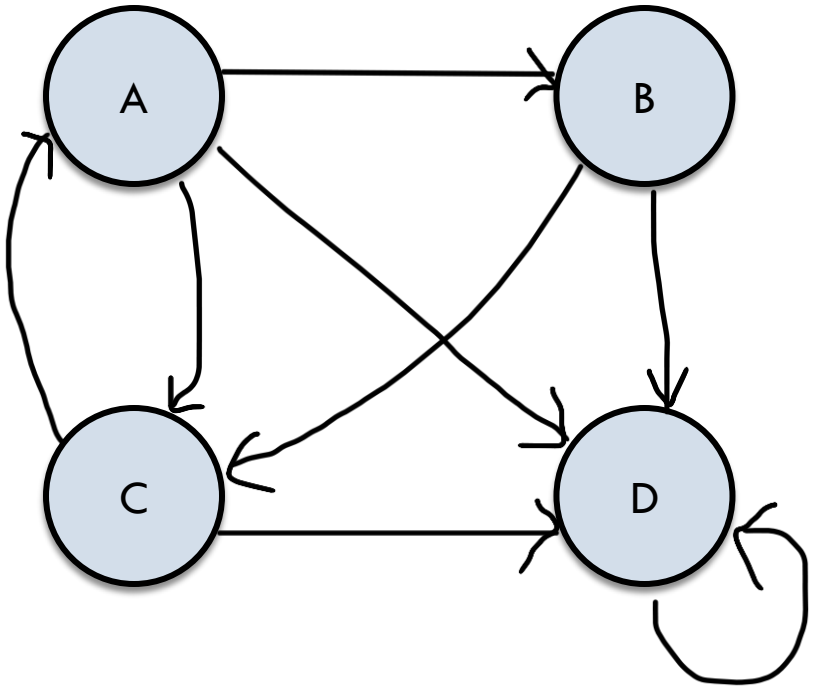
$$h = [h(1), h(2), \ldots, h(n)]T,$$

- Then,

$$a = L^T h$$

$$h = La$$

How to compute hub and authority scores



Adjacency Matrix

L =

	A	B	C	D
A	0	0	1	0
B	1	0	0	0
C	1	1	0	0
D	1	1	1	1

We know that:

Authority score $a = L^T h$

Hub Score $h = La$

How to compute hub and authority scores

- Let us assume that initially, all the nodes have equal hub score of 1.
- So, $\mathrm{h} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ we already have $L = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$
- $L^T = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- Authority score $\mathrm{a} = \mathrm{L}^{\mathrm{T}}\mathrm{h}$

How to compute hub and authority scores

- **Iteration 1:-**
- Authority score $a = L^T h$
- $a = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 2 \\ 1 \end{bmatrix}$
- Normalization of authority score:

  divide each term by $\sqrt[2]{(3^2 \quad +2^2 + \quad 2^2 + \quad 1^2)} = \sqrt{18} = 4.24$
- $a = \begin{bmatrix} 3/4.24 \\ 2/4.24 \\ 2/4.24 \\ 1/4.24 \end{bmatrix} = \begin{bmatrix} 0.707 \\ 0.472 \\ 0.472 \\ 0.236 \end{bmatrix}$

- Hub Score $h = La$
- $h = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0.707 \\ 0.472 \\ 0.472 \\ 0.236 \end{bmatrix} = \begin{bmatrix} 0.472 \\ 0.707 \\ 1.179 \\ 1.887 \end{bmatrix}$
- Normalization of hub score:

divide each term by $\sqrt[2]{(0.472^2 \quad +0.707^2 + \quad 1.179^2 + \quad 1.887^2)} = \sqrt{5.673} = 2.382$

$$h = \begin{bmatrix} 0.472/2.382 \\ 0.707/2.382 \\ 1.179/2.382 \\ 1.887/2.382 \end{bmatrix} = \begin{bmatrix} 0.198 \\ 0.297 \\ 0.495 \\ 0.792 \end{bmatrix}$$

- So, at the end of iteration 1:
- Authority scores of nodes are: $\begin{bmatrix} 0.707 \\ 0.472 \\ 0.472 \\ 0.236 \end{bmatrix}$
- Hub scores of nodes are: $\begin{bmatrix} 0.198 \\ 0.297 \\ 0.495 \\ 0.792 \end{bmatrix}$

- **Iteration 2:-**
- Authority score $a = L^T h$
- $a = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0.198 \\ 0.297 \\ 0.495 \\ 0.792 \end{bmatrix} = \begin{bmatrix} 1.584 \\ 1.287 \\ 0.99 \\ 0.792 \end{bmatrix}$
- Normalization of authority score:

  divide each term by $\sqrt[2]{(1.584^2 + 1.287^2 + 0.99^2 + 0.792^2)} = 2.4$

$$a = \begin{bmatrix} 1.584/2.4 \\ 1.287/2.4 \\ 0.99/2.4 \\ 0.792/2.4 \end{bmatrix} = \begin{bmatrix} 0.66 \\ 0.53 \\ 0.41 \\ 0.33 \end{bmatrix}$$

Hub Score h = La

$$h = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} . \begin{bmatrix} 0.66 \\ 0.53 \\ 0.41 \\ 0.33 \end{bmatrix} = \begin{bmatrix} 0.41 \\ 0.66 \\ 1.19 \\ 1.93 \end{bmatrix}$$

Normalize the hub score in the similar manner

Final values of a and h after second iteration are:

$$a = \begin{bmatrix} 0.66 \\ 0.53 \\ 0.41 \\ 0.33 \end{bmatrix} \quad h = \begin{bmatrix} 0.171 \\ 0.275 \\ 0.496 \\ 0.805 \end{bmatrix}$$

Internet Mathematics and Future Directions

Internet Mathematics, Future directions

- The development of World Wide Web lead to a whole new field of mathematics called Internet Mathematics. It is a mixture of **probability, linear algebra, graph theory and dynamical systems**, designed to answer stringent problems about how information propagates over the net.
- From a web designer's perspective, it is important not only to create a nice web site, but it is important what other pages link to your web site. A good PageRank can turn your business into a very profitable one.
- From Google's point of view however, it is important to keep PageRank computations accurate, and to prevent **fraud attempts**.
- ***Link spam*** is defined as the intentional attempt to improve the ranking of a page on the search engine, by increasing the number of pages that link to it.
- ***Link farms*** involve creating strongly connected communities of web pages referencing each other, humorously known as "mutual admiring societies".

Internet Mathematics, Future directions

- Another research direction is the analysis and comparison of different ranking algorithms.
- One property that makes an algorithm good is stability.
- The World Wide Web is in constant change, pages are added and deleted at every moment. Stability measures how much rankings change if the Web graph is slightly perturbed.
- Ideally, we would like the ranking algorithms to be stable so that, for instance, we would not have to constantly recompute the PageRank.

Internet Mathematics, Future directions

- Although in our previous lectures we only talked about finite graphs, we can also view the web as an infinite graph.
- The reason for this is that web pages are of two kinds: static and dynamic.
- Static pages contain information that remains unmodified.
- Dynamic pages are those built "on the spot", as a result of a query.
- For instance, if we do a Google Search with the phrase "mathematics and computer science", Google returns a page with relevant links and their descriptions.
- Of course, Google does not have a page prepared beforehand, for every possible query a random user might have.

Internet Mathematics, Future directions

- For every query it receives, it searches through its index to track the documents which contains the given search words, then Google runs the Page Rank algorithm to rank the relevant hits.
- Only after that, does it build up the answer to the user.
- We call such a page a dynamical page. While the number of static pages is finite (over 30 billion, but still finite), the number of dynamical pages is virtually infinite.
- Such a complete view of the web including both static and dynamic content can be given by infinite graph theory.
- The mathematics behind it is both interesting and challenging, and it is an ongoing research field.

SALSA: The Stochastic Approach for Link-Structure Analysis

SALSA: The Stochastic Approach for Link-Structure Analysis

- The paper# introduces **SALSA**, a stochastic algorithm designed to identify **authoritative Web pages** by analyzing the link-structure of the Internet.
- This paper was build upon Jon Kleinberg's "hubs and authorities" model (HITS algorithm), the authors utilize **random walks on bipartite graphs** to rank page relevance for broad-topic queries.
- They demonstrated that while SALSA follows a similar meta-algorithm to Kleinberg's Mutual Reinforcement approach, it is **computationally more efficient** because it effectively functions as a weighted in-degree analysis.
- Crucially, the authors identify the **TKC (Tightly Knit Community) Effect**, a topological phenomenon where small, highly interconnected clusters can distort search rankings.
- Their findings suggest that **SALSA is more robust** than previous models, as it successfully resists the TKC Effect to provide more diverse and accurate results.
- Consequently, the study advocates for integrating **stochastic processes** to enhance the precision of modern search engines.

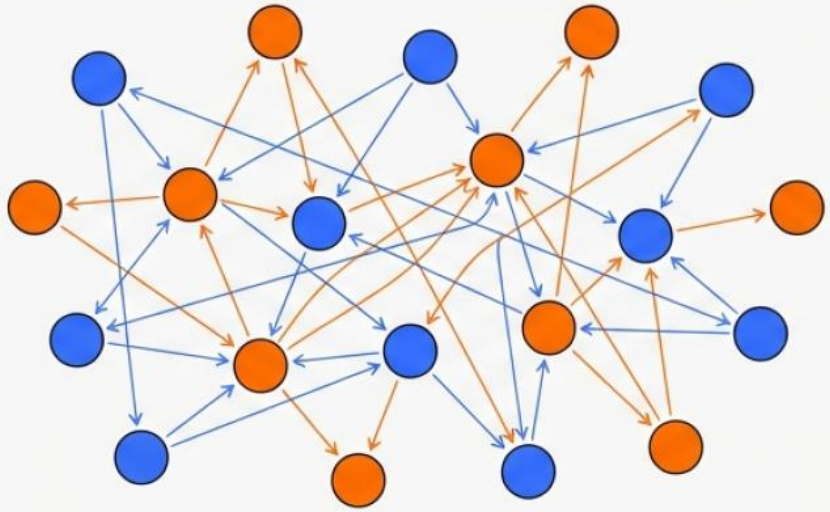
Lempel, R., & Moran, S. (2001). SALSA: the stochastic approach for link-structure analysis. *ACM Transactions on Information Systems (TOIS)*, *19*(2), 131-160.

SALSA: The Stochastic Approach for Link-Structure Analysis

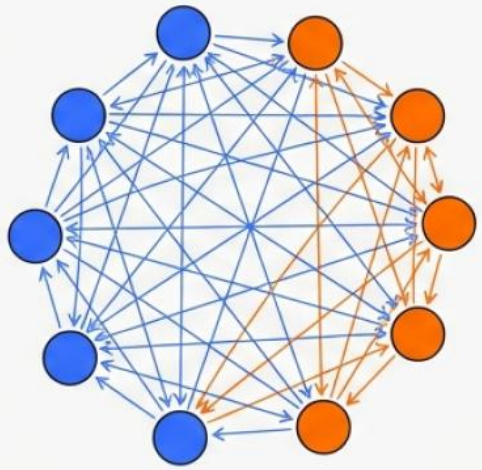
The Vulnerability: When Communities Become Cliques

The Tightly Knit Community (TKC) Effect

Healthy Community



The TKC (Tightly Knit Community)



The Flaw: HITS is attracted to the DENSEST part of the graph. A small, highly **interconnected** set of pages (like a spam ring) can **score artificially high, overriding the actual relevant community**.

SALSA: The Stochastic Approach for Link-Structure Analysis

Case Study: The 'Movies' Query Anomaly

Query: "movies"

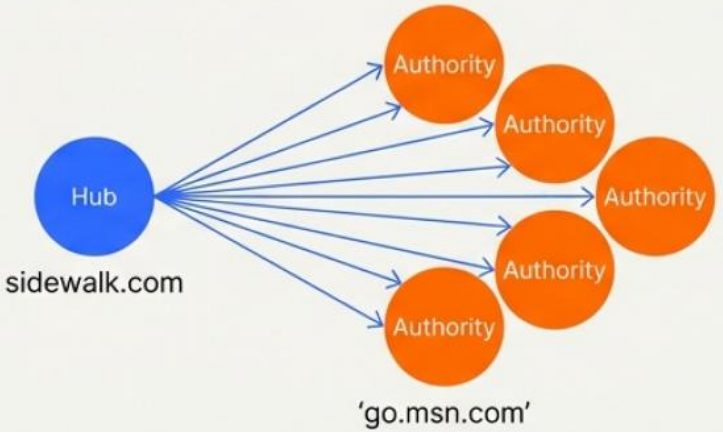
Algorithm: Kleinberg's HITS

Result: The algorithm was hijacked by a navigational structure within a single domain network, completely missing IMDb, Disney, and Hollywood.com.

HITS Top Authorities Results

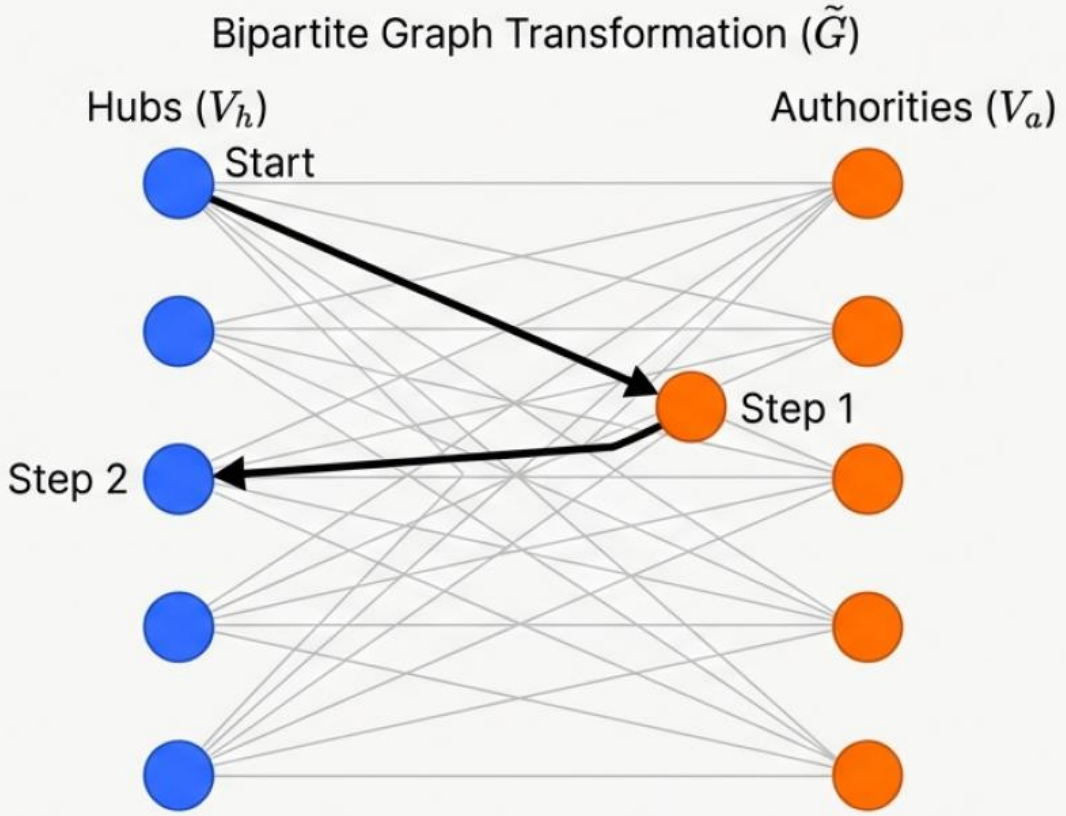
Authority Page	Score
1. go.msn.com/npl/msnt.asp	0.1673
2. go.msn.com/bql/whitepages.asp	0.1672
3. go.msn.com/bsl/webevents.asp	0.1672
... (Top 30 results were all go.msn.com sub-pages)	0.1672

Near-identical scores indicate a manufactured clique (TKC).



SALSA: The Stochastic Approach for Link-Structure Analysis

Decoupling the Web: The Bipartite Graph Walk



The SALSA Random Walk is a two-step process:

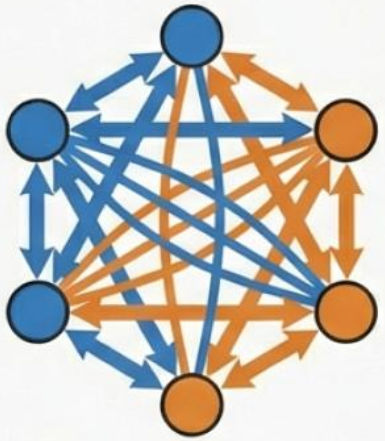
1. Hub → Authority
2. Authority → Hub

This creates two distinct Markov chains—one for Hubs and one for Authorities—decoupling the rigid link that causes the TKC effect.

SALSA: The Stochastic Approach for Link-Structure Analysis

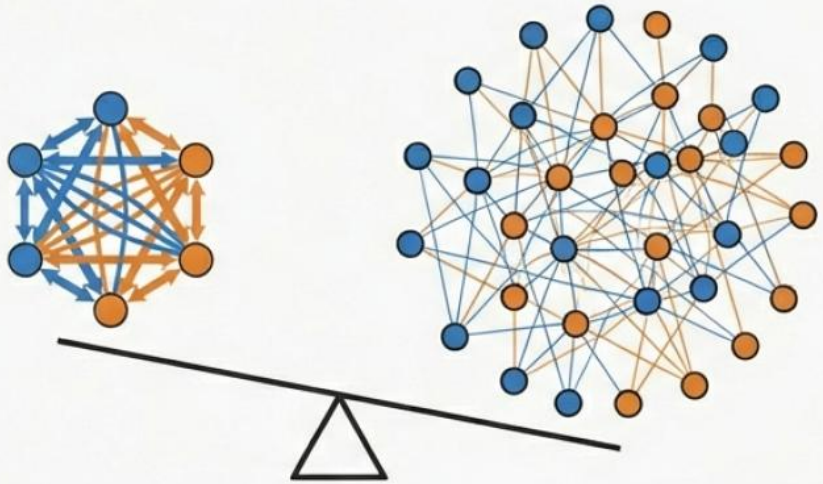
Anatomy of the Escape: Why SALSA Resists the TKC

HITS: Density Matters



A clique of size *k* produces weights proportional to link DENSITY.

SALSA: Size Matters



In a Random Walk, authority is weighted by the SIZE of the component. A small spam ring gets a low global score. A large organic community gets a high global score.

More Network Measures

Transitivity and Reciprocity

- **Transitivity:**
- In abstract mathematics, transitivity in a relation means that when there exists a tie from entity a to entity b, and also from i to c, then there also exists a tie from a to c.
- The same analogy is adopted in the theory of social network as follows: "Friends of one's friends are her friends as well".
- Formally, in network G(V, E), when there is an edge between nodes v_1 and v2, and also an edge between v2 and v3. if we find another edge between v1 and v3 in G, then we observe a transitive linking in G.

Transitivity and Reciprocity

- Clearly, transitivity involves three edges.
- These three edges along with their corresponding end nodes form a triangle or a triad.
- The presence of such triangles provides a measure of transitive behaviour in the network.
- More the triangles in the network, higher is its transitivity.
- Transitivity in a network may be measured using clustering coefficient discussed in earlier lecture.
- the same may be applied to directed networks as well, where instead of triangles, we consider the loops of size 3 (or motifs)

Transitivity and Reciprocity

- **Reciprocity:**
- Reciprocity is a measure of the likelihood of nodes in a directed network to be linked to each other.
- Formally, given a directed edge from node u to node v, if we find another directed edge from v to u in the network G, then we notice an instance of reciprocal behaviour in G.
- Informally, reciprocity refers to "If you would follow me, I shall follow you back.“
- To measure reciprocity of a network G(V, E), we first count the number of reciprocal pairs and then normalize the same by maximum possible pairs in the network.

Transitivity and Reciprocity

A directed network can have at most |E| / 2 reciprocal pairs. Then the Reciprocity R of a directed network G is given as:

$$R = \frac{2}{|E|} \sum_{i,j;\, i<j} a_{ij} \times a_{ji}$$

where $A = (a_{ij})$ is the adjacency matrix of G.

On simplification R reduces to:

$$R = \frac{2}{|E|} \times \frac{1}{2} Tr(A^2) = \frac{1}{|E|} Tr(A^2)$$

Where Tr(X) refers to the trace of a matrix X, which is the sum of diagonal elements of X.

Structural Equivalence

In the structural similarity, we look at the nodes which are in the common neighbourhood to the given two nodes. For example, in a network of your friends, you and your best friend might be most similar since you may share a lot of common friends with your best friend. Let a and b be two nodes in the network G. Let $N(a)$ and $N(b)$ be the neighbourhoods of nodes a and b, respectively. Different measures through which we can measure structural similarity between nodes a and b are as follows.

Structural Equivalence

1. **Common neighbours:** It is simply the number of common neighbours shared in the neighbourhoods $N(a)$ and $N(b)$. Mathematically, it is defined as,

$$\sigma_{CN}(a, b) = |N(a) \cap N(b)| \tag{2.16}$$

2. **Jaccard similarity:** Here we simply normalise common neighbours by the combined size of the neighbourhoods of the two nodes, $|N(a) \cup N(b)|$. Mathematically, it is defined as,

$$\sigma_{JS}(a, b) = \frac{|N(a) \cap N(b)|}{|N(a) \cup N(b)|} \tag{2.17}$$

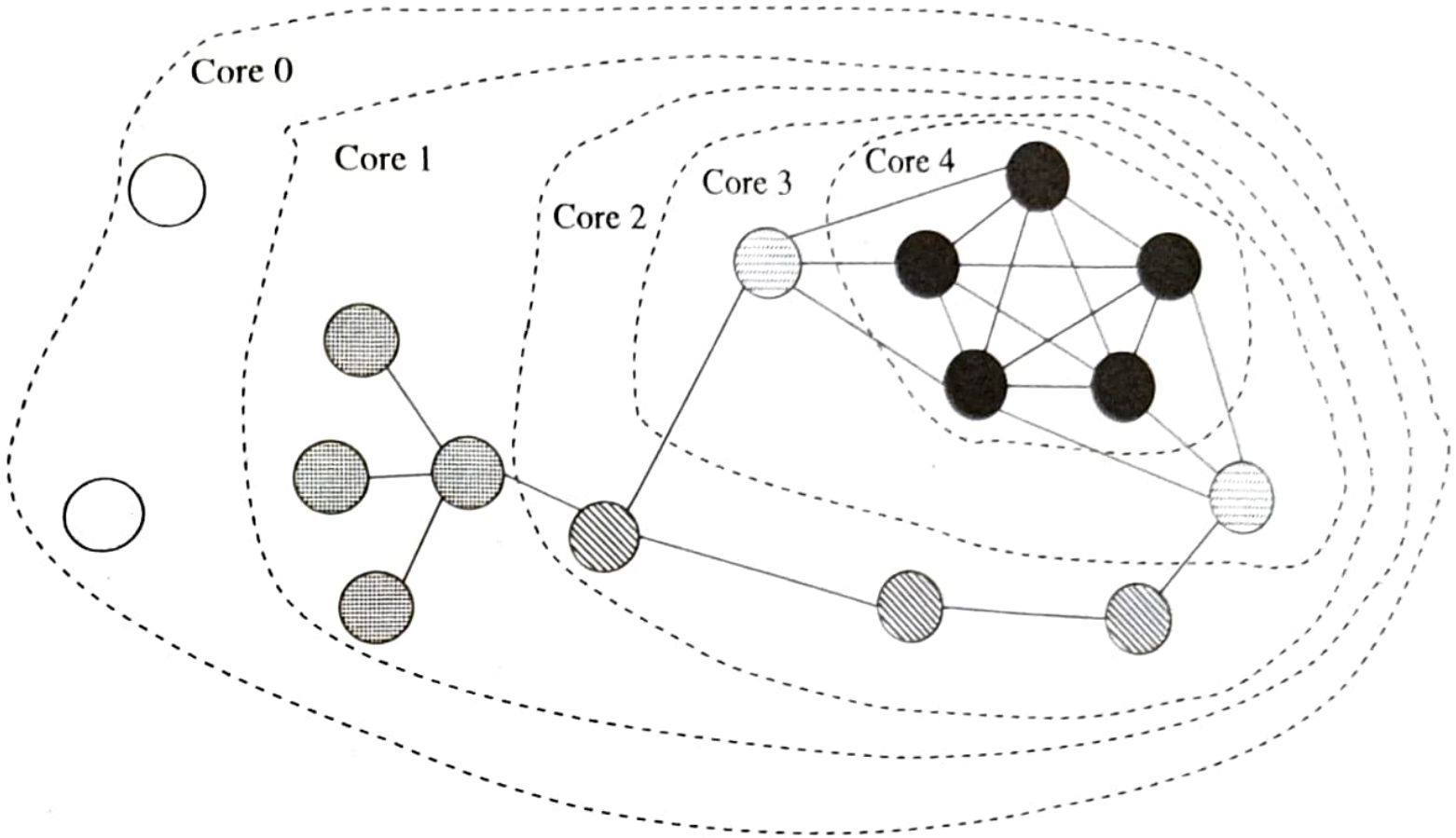
3. **Cosine similarity:** Here, we normalise the common neighbours by the individual sizes of the neighbourhoods, $|N(a)|$ and $|N(b)|$.

$$\sigma_{CS}(a, b) = \frac{|N(a) \cap N(b)|}{\sqrt{|N(a)||N(b)|}} \tag{2.18}$$

Degeneracy

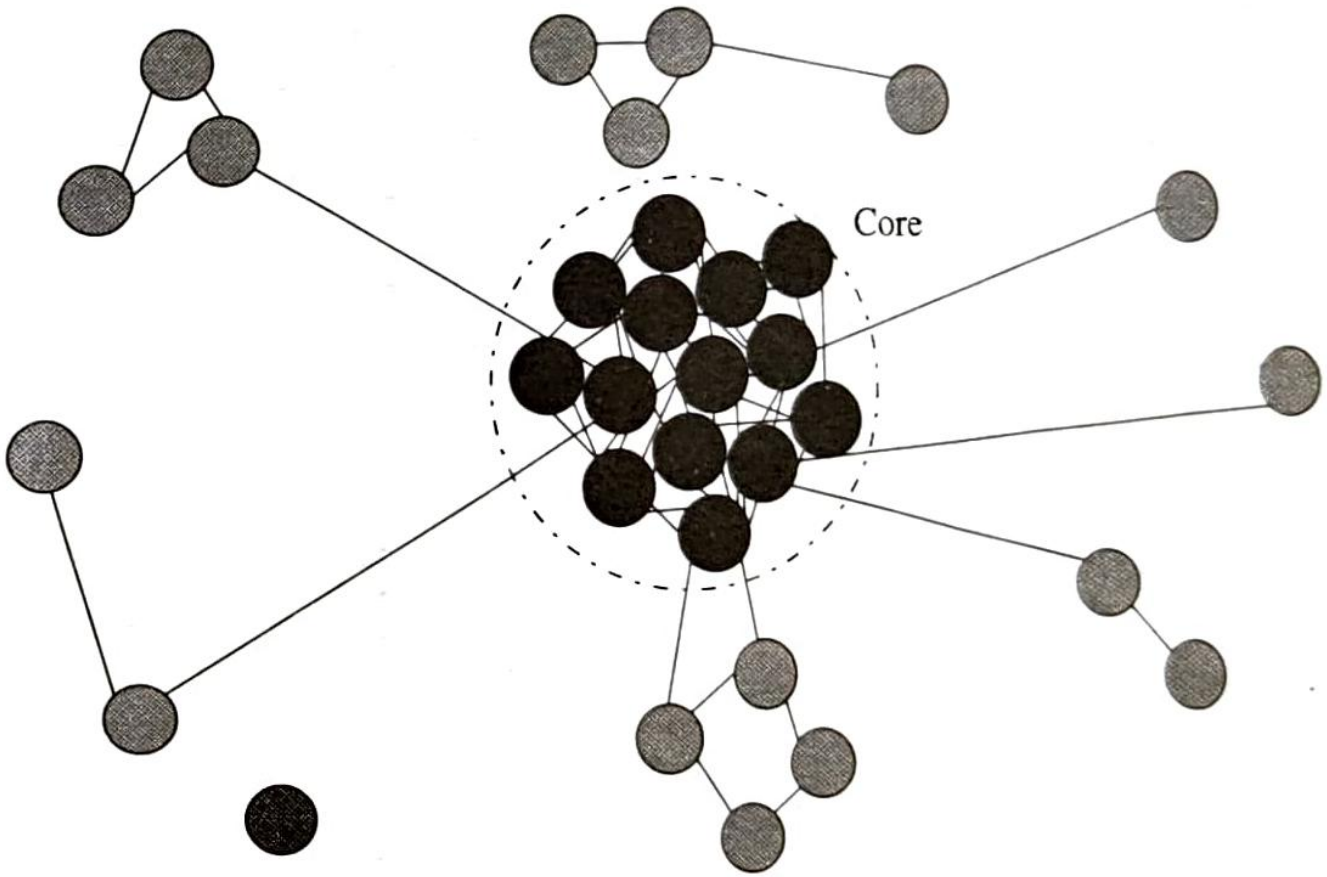
- Network degeneracy is an indirect measure of the sparsity of the network.
- **K-Core:**
- For an undirected network G. the k-core is the maximal subgraph of G in which every node has at least & neighbours for are of degree at least k).
- How would we construct the k-core of a network G?
- The first thought that may come to our mind i to simply drop all the nodes with degree less than k.
- dropping all nodes of degree of (undropped) neighbours to thereby making them ineligible to be part of the k-core subgraph.
- Therefore, to form the k-core subgraph we recursively "shave" nodes with degree less than k.
- Figure on next slide shows k-core subgraphs for k = 0, 1, 2, 3, 4 for a network.

Degeneracy



K-core subgraphs for k = 0, 1, 2, 3, 4 for a network

Degeneracy

- **Coreness:**
- The coreness or core number of a node is the order of the highest order core that the node belongs to.
- A node has a coreness k in network G if it belongs to the k-core subgraph, but does not belong to the (k+1)-core subgraph of G.
- For example, in Figure shown in next slide, nodes inside the central most 4-core subgraph have coreness 4.
- Coreness is similar to centrality in a sense that it also measures the most significant or prestigious nodes in the network.
- This is based on the intuition that nodes with higher coreness tend to be more central and connected and hence more influential in the network.

Degeneracy